

Day 89 D/E: OpenGL and BGFX Renderer Boundary

VSEPR-SIM Work Order Authority

Day 89

Consolidated scope

This record combines Day 89-D, OpenGL scale consumption, and Day 89-E, BGFX scale consumption, at their shared visual-data boundary. Native CPU/GDI remains Day 89-F. The scientific producer and artifact contract are authoritative; neither renderer may add, mutate, or silently reduce scientific rows.

The input package is:

```
vsepr.dual_backend_3d.v1  
manifest.json, scenes.csv, and particles.csv
```

Each consumer must preserve scene identity, particle identity, positions, radii, state values, bounds, and declared particle count. Presentation-specific camera, projection, shading, and frame pacing remain downstream concerns.

Current evidence

The shared stress package is available under `out/day89_recheck`. Its manifest contains five scenes and 1,442 particles, and the machine validator passes. This proves the upstream package boundary, not renderer parity.

The remaining D/E risks are explicit. The OpenGL visual dependency path is not currently a fresh verified build in this checkout. The BGFX consumer still needs deterministic malformed-input and invalid-scene diagnostics, and its current path is debug-text based rather than a validated visible-geometry path. No machine-checked OpenGL/BGFX count, bounds, and state parity result has yet been recorded.

Acceptance gate

Day 89-D/E is complete only when both consumers, or an explicit capability blocker for either, are recorded against the same fresh package and:

1. load the manifest and selected scene without uncaught parse or selection exceptions;
2. report the declared scene and particle counts;
3. preserve the same particle bounds, positions, and state values used by the neutral artifact;
4. emit deterministic diagnostics for missing, empty, malformed, and invalid-scene inputs;
5. demonstrate that renderer execution does not mutate the scientific CSV or manifest; and
6. provide a machine-readable parity result or a bounded, reproducible blocker.

Status

Day 89-D/E is **TODO / ACTIVE NEXT**. The combined authority closes the numbering gap and gives OpenGL and BGFX one contract, but it does not claim implementation completion. Day 89-F may proceed against the same neutral package while D/E remains open; Day 89-G consumes the three renderer results.